Q1 Given three integers n, a and b. A number is magical no. they are magical if it is divisible by either a or b Since, the answer can be large use $10^9 + 7$.

eg ⇒ n = 1, a = 3, b = 2

ALGORITHM :->

1.) Compute the GCD
→ find GCD using Euclich Algorithm

2.) Compute the LCM
→ Since $a \times b = LCM \times GCD \Rightarrow LCM = (a/GCD) * b$

3.) Define Search space

low = min (a, b);
High = n × min (a, b);

∴ because n × (min (a, b)) will for sure be 7 * special No.

4.) Binary Search

while (l < r) {
mid = l + (r - l)/2;

Compute $\left(\frac{mid}{a} + \frac{mid}{b} - \frac{mid}{LCM} \right)$

T.C → log n
S.C ⇒ O(1)

perform Binary Search
if Count > n {
r = mid - 1
else

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5) Print ans as the rth special number if we get Count == n.

```
int rth special (int a, int b, int c) {
    long LCM = (a / gcd (a,b)) * b;

    int l = min (a, b);
    int r = r * min (a, b);
    int ans = -1;

    while (l ≤ r) {
        int mid = l + (r-l)/2

        if (valid ( mid , a, b, LCM , n)) {
            ans = mid;
            r = mid -1;
        else
            l = mid + 1

return ans;

boolean valid (int mid, int a, int b, int LCM, int n) {
    long count = mid/a + mid ( l - mid / LCM;
    return Count >= r;

int GCD (int a, int b) {
    return b==0 ? a : GCD (b, a % b);
```